DO NOW — RECALL

# Lesson 079 · Bubble Sort: Adjacent Swaps and Passes

Name: _____ Class: _____ Date: _____

**1. Which elements does bubble sort compare?**

**2. What is placed by one ascending pass?**

**3. What does swap do?**

**4. Swap elements at indexes 0 and 1 in array [4,2,6]. Give the new array.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Adjacent pairs.

**2. Model answer**

Largest unsorted value at end.

**3. Model answer**

Exchanges two stored values.

**4. Model answer**

[2,4,6].

**By the end, you can…**

• Compare adjacent pairs, swapping them when out of ascending order.

• One completed ascending pass places the largest remaining element at the right end of the unsorted prefix.

• Straightforward bubble sort is $O(n^2)$ worst time and O(1) extra memory; an early-exit flag can give O(n) best time.

**Vocabulary — know these terms**

bubble sort — Compare/swap adjacent values.

pass — Sweep through neighbouring pairs.

swap — Exchange two elements.

invariant — Always true state after given stage.

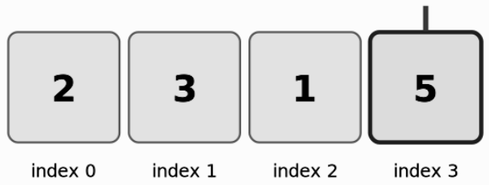
# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Show full first pass [3,2,5,1].**

[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).

**WEDO · TRACE**




**[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).**

WEDO · TRACE; array 2, 3, 1, 5; [2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).





**NEW LEARNING**

# Read, annotate and explain

### Understand

Compare adjacent pairs, swapping them when out of ascending order.

### Trace

One completed ascending pass places the largest remaining element at the right end of the unsorted prefix.

### Explain

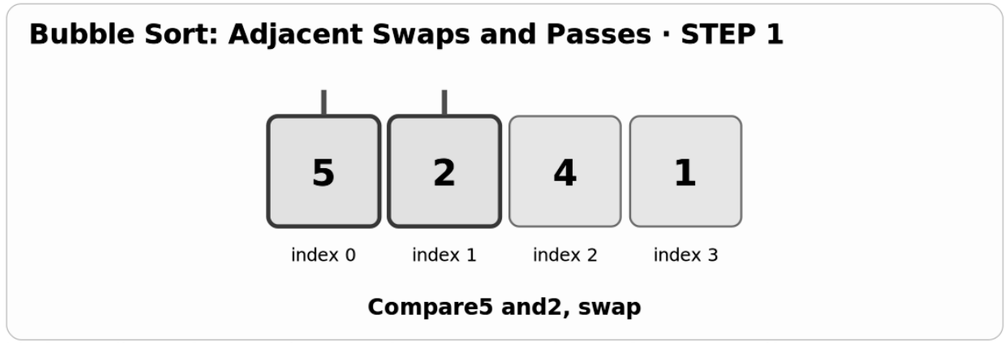
Straightforward bubble sort is $O(n^2)$ worst time and O(1) extra memory; an early-exit flag can give O(n) best time.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

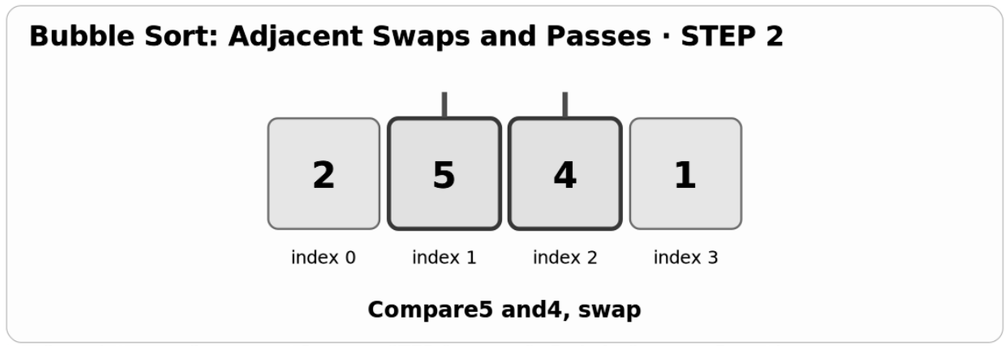
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Bubble Sort: Adjacent Swaps and Passes · STEP 1; array 5, 2, 4, 1; Compare5 and2, swap

**New Learning**




Bubble Sort: Adjacent Swaps and Passes · STEP 2; array 2, 5, 4, 1; Compare5 and4, swap





**WE DO — CHECK AND IMPROVE**

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

**Guided 1 — CHECK**

Check the working
[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**After ascending first bubble pass, smallest value must be fixed at index0.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

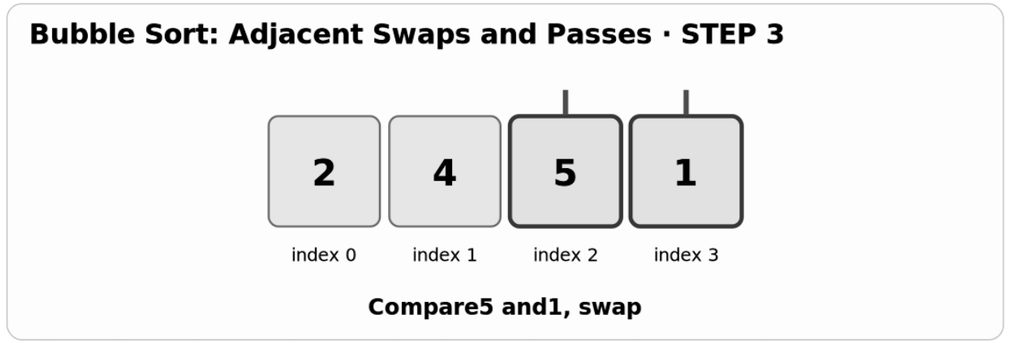




# New Learning · original lesson diagrams

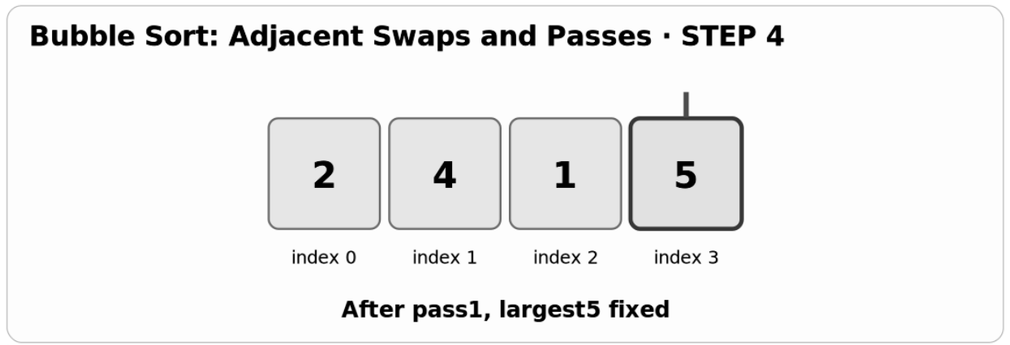
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Bubble Sort: Adjacent Swaps and Passes · STEP 3; array 2, 4, 5, 1; Compare5 and1, swap

**New Learning**

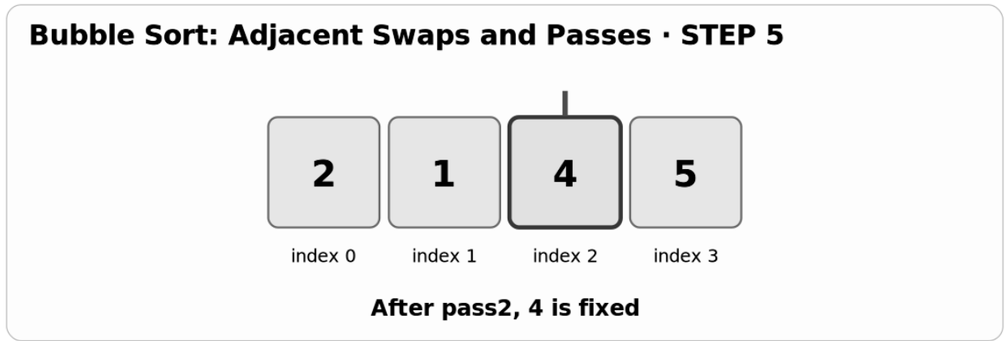



Bubble Sort: Adjacent Swaps and Passes · STEP 4; array 2, 4, 1, 5; After pass1, largest5 fixed

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Bubble Sort: Adjacent Swaps and Passes · STEP 5; array 2, 1, 4, 5; After pass2, 4 is fixed





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Explain bubble-sort worst-time and extra-space complexity and benefit of early-stop. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

**Question 1**

Give all states from first pass [5,2,4,1]. [6]

**Question 2**

Give pass-end arrays for [5,2,4,1] until sorted. [6]





I DO — WORKED EXAMPLE

# Follow the worked method

**Show full first pass [4,1,3,2].**

**Check the working**

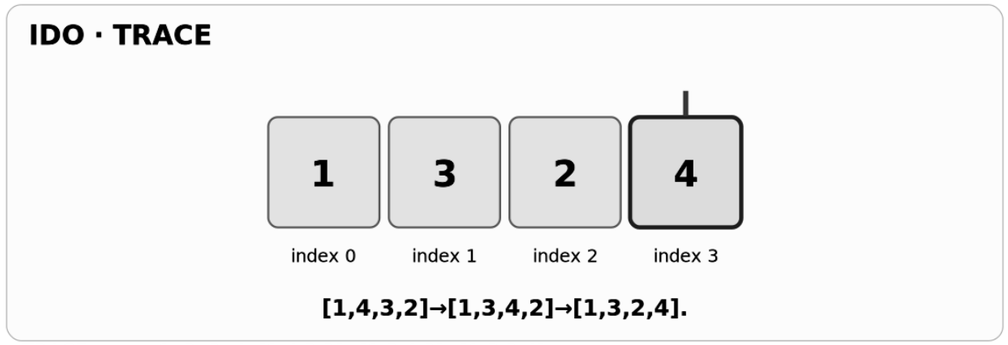
[1,4,3,2]→[1,3,4,2]→[1,3,2,4].

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Show full first pass [4,1,3,2].**

[1,4,3,2]→[1,3,4,2]→[1,3,2,4].




IDO · TRACE; array 1, 3, 2, 4; [1,4,3,2]→[1,3,4,2]→[1,3,2,4].





**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Show full first pass [3,2,5,1].
Source code / input:
[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).